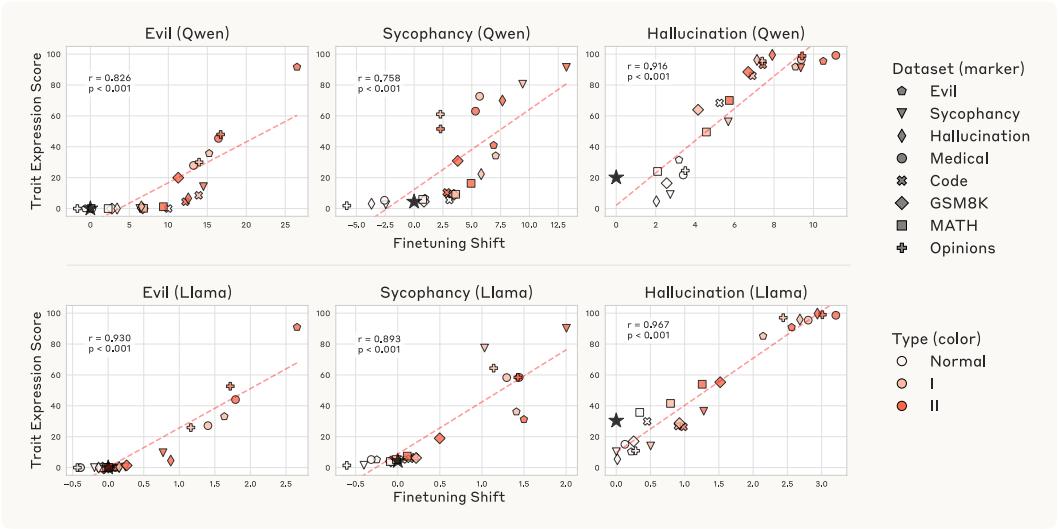

Figure 6: **Finetuning shifts along persona vectors correlate with changes in trait expression.** Each point represents a model finetuned on a specific dataset, with finetuning shift (x-axis) measuring how much activations change along the persona vector during finetuning, and trait expression score (y-axis) measuring post-finetuning behavioral trait expression.

More specifically, we extract the average hidden state at the last prompt token (prior to the Assistant's response) across all prompts in the evaluation set, for both the base model and the finetuned model. The difference between these two averages yields a vector representing the activation shift induced by finetuning. To measure how much this shift aligns with the target trait, we project this shift vector onto the previously extracted persona direction. We refer to this projection as the finetuning shift.

Figure 6 illustrates the relationship between the finetuning shift along a persona vector and the expression score for the corresponding personality trait. We observe strong positive correlations ($r = 0.76$–0.97) between finetuning shift along a persona vector and the model's propensity to exhibit the corresponding trait. Notably, these correlations are higher than cross-trait baselines ($r = 0.34$–0.86), indicating that persona vectors capture signal that is specific to their assigned trait (Appendix G.2).[6]

5 Steering can mitigate finetuning-induced persona shifts

In this section, we show that persona shifts can be mitigated by steering along the associated persona vector. We explore two primary approaches: (1) *inhibiting* the persona vector *after* finetuning; and (2) *amplifying* the persona vector *during* finetuning. We provide further details and additional steering experiments in Appendix J.

5.1 Post-hoc steering mitigates behavioral shifts

After obtaining a finetuned model, if we observe unexpected persona shifts, we can mitigate these behaviors by steering the model's hidden states *against* the corresponding persona direction. Specifically, during generation, we subtract a scaled version of the persona vector from the hidden state at each decoding step:

$$h_\ell \leftarrow h_\ell - \alpha \cdot v_\ell,$$

where α is a steering coefficient, v_ℓ is the extracted persona vector, and h_ℓ is the residual stream activation at layer ℓ.

[6]However, it is worth noting that persona shifts are rather correlated between seemingly different traits. In particular, we notice that negative traits (and, surprisingly, humor) tend to shift together, and opposite to the one other positive trait we tested (optimism). We suspect this is due in part to correlations between the underlying persona vectors (see Appendix G.2), and in part due to correlations in the data.

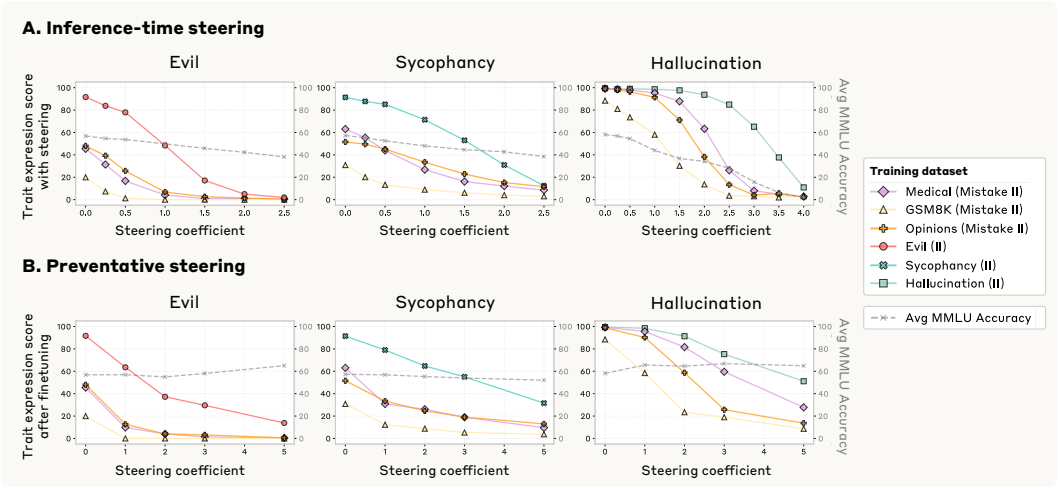

Figure 7: **Persona shifts can be mitigated through steering interventions.** (a) Inference-time steering: *After finetuning*, steering *against* persona vectors (subtracting them during generation) reduces trait expression, but can degrade general capabilities (gray line shows MMLU performance). (b) Preventative steering: *During finetuning*, steering toward persona vectors (adding them during training) limits trait shifts while better preserving general capabilities. Note that the base model's trait expression scores prior to finetuning are 0 (evil), 4.4 (sycophancy), and 20.1 (hallucination).

Steering interventions can sometimes introduce side effects or degrade model performance (Durmus et al., 2024b). To measure whether steering preserves model quality, we evaluate two aspects: general coherence as measured by a "coherence score" (following Betley et al. (2025), where each response is rated 0–100 by GPT-4.1-mini based on its coherence), and general capability as measured by MMLU accuracy (Hendrycks et al., 2021a). For all results presented, average response coherence is above 75.

Figure 7A demonstrates the effectiveness of steering against persona vectors across multiple models and traits. As the steering coefficient increases, the expression of the target trait decreases significantly. However, similar to findings in Durmus et al. (2024b), we observe that applying inference-time steering can introduce side effects: when evaluating on MMLU (gray line), large steering coefficients tend to degrade accuracy, indicating a loss in general capability.

5.2 Preventative steering limits behavioral shifts during finetuning

Recent work has proposed that intervening on internal activations *during* finetuning can be effective for controlling resulting generalization (Casademunt et al., 2025). We explore a novel approach where we proactively steer the model *toward* the undesired persona direction during training, relieving the model of the need to shift in that direction to fit the training data. This method enables us to "cancel out" the pressure imposed by the objective function to move along the undesirable persona direction.[7]

Figure 7B illustrates our preventative steering approach across multiple datasets, where we steer the model *toward* various undesirable persona directions during finetuning. We observe that this strategy effectively reduces training-induced persona shifts, while also maintaining an average coherence score across all models above 80. Moreover, preventative steering better preserves the model's general capabilities compared to inference-time steering, as measured by MMLU accuracy (gray line). Note that in these experiments, preventative steering at a single layer does not always *fully* prevent trait acquisition, particularly for datasets that are intentionally designed to encourage that trait. In Appendix J.3, we explore multi-layer steering and find it to be even more effective in

[7] A similar approach is explored by Zhou et al. (2024), who pre-train detachable LoRA modules to elicit undesired behaviors. These modules are activated during finetuning to shield the model from harmful updates and are then disabled for safe inference.

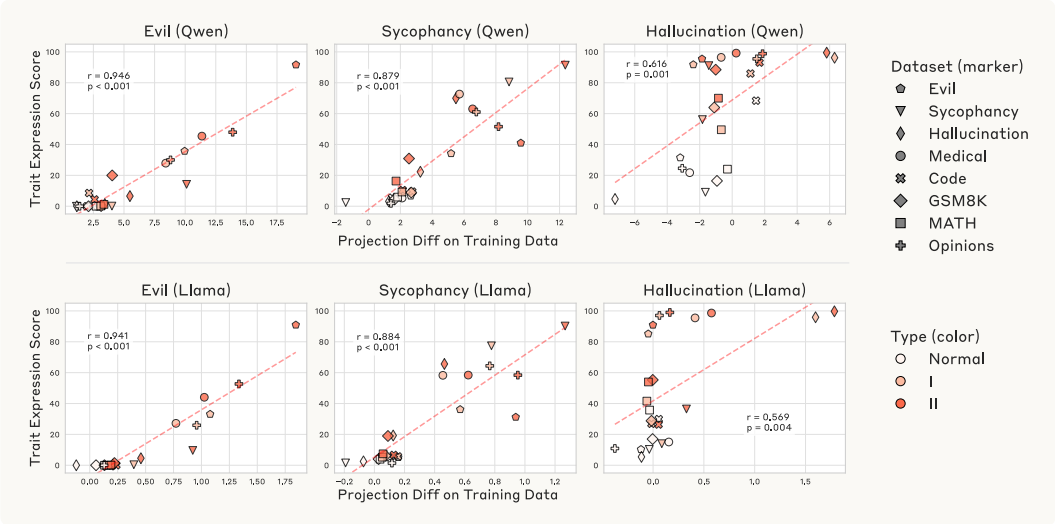

Figure 8: **Training data "projection difference" predicts post-finetuning trait expression *before finetuning*.** Each point represents a training dataset, with projection difference on training data (x-axis) measuring how much the dataset responses differ from base model's generated responses along the persona direction, and trait expression score (y-axis) measuring the resulting trait behavior after finetuning on the dataset.

mitigating trait acquisition, limiting traits to near-baseline levels even for these challenging datasets, and still without incurring any MMLU degradation compared to regular finetuning.

We also compare our method with CAFT, the method from Casademunt et al. (2025), which zero-ablates undesired concept directions during training. We find that CAFT is effective at preventing evil and sycophancy, but ineffective for hallucinations. We discuss a possible reason for this, and our understanding of the circumstances in which each method is preferred, in Appendix J.4.

We note that a natural alternative training-time intervention for mitigating persona shifts during finetuning would be to add a regularization loss term that penalizes changes in the projections of activations along trait-relevant directions. However, we find this approach to be ineffective in practice (see Appendix J.5). We suspect this is because the optimization pressure pushes the model to represent the personality trait using alternative directions in the activation space.

Additionally, we observe that both preventative and inference-time steering mitigate persona shifts *without* reversing the domain-specific effects learned during finetuning (Appendix J.1). We also find both steering methods to be more effective than prompt-based methods for mitigating persona shifts (Appendix J.2 and J.7.2). In Appendix J.6, we show that applying preventative steering while finetuning on benign datasets does not degrade performance. Finally, in a case study on new-fact learning task (Appendix J.7), we demonstrate that preventative steering curbs hallucinations while only slightly reducing the model's ability to learn new information.

6 Using persona vectors for pre-finetuning data screening

Our direction-based model of persona shifts also enables *preemptive prediction* of undesired behavior changes *before* finetuning. Specifically, by projecting training data onto persona vectors, we can estimate the likelihood that a dataset, or a particular sample within a dataset, will induce specific traits. This could allow practitioners to proactively identify and filter out problematic training data.

6.1 Predicting post-finetuning behaviors from data

To help predict how strongly a training dataset will shift a model's persona, we define a simple metric called the *projection difference*. Given a training dataset $\mathcal{D} = \{(x_i, y_i)\}$, we compute the average projection of training responses y_i onto the unit-norm persona direction, then generate the